Nonlocal Boundary Conditions for Split-Step Padé Approximations of the Helmholtz Equation With Modified Refractive Index

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***Abstract*—This research is concerned with the numerical methods for the electromagnetic wave propagation over the Earth's surface. Discrete nonlocal boundary conditions for the Padé parabolic equation are derived. A linear dependence of the modified refractive index on the height outside the computational domain can be taken into account. Thus, the earth-flattening transformation can be used. The proposed approach gives an opportunity to truncate the integration domain without introducing an artificial absorbing layer in the vicinity of the upper boundary. Padé approximations of the exponential propagation operator allow one to carry out the calculations with a rather large step along the longitudinal coordinate. The elaborated method can utilize the two-way parabolic equation approach for scattering problems. Comparison with the results obtained by the split-step Fourier method is given.**

***Index Terms*—Helmholtz equation, nonlocal boundary condition (NLBC), Padé approximations, parabolic equation (PE), radio-wave propagation.**

I. Introduction

PERFORMANCE of radars and various wireless communication systems can be significantly influenced by the environment. Abnormal tropospheric refractive index and irregular terrain may have a significant impact on the propagation of electromagnetic waves. The effects of tropospheric propagation can significantly increase the zone of radio visibility, but also provoke the existence of the skipping zones. Hence, the numerical models that can incorporate the refractivity and terrain factors can be useful for design and operation of various radio systems.

The most common approach to provide the solution of the above-mentioned class of problems is the parabolic equation (PE) method [1], [2]. The numerical algorithms for the PE include the two widely used methods: split-step Fourier (SSF) method [3] and Crank–Nikolson finite difference scheme [4]. The main advantage of the SSF method is the ability to use sufficiently large distances between the nodes of the computation grid, thus achieving a high performance. The main disadvantage is the laborious accounting of boundary conditions [5]. Finite-difference PE approximations also provide a good computational performance along with a more straightforward accounting of the lower impedance boundary condition. Significant feature of the considered problem lies in the fact that the computations should be performed in a semi-infinite domain. Thus, it is necessary to introduce the truncation procedure in order to get rid of false reflections of the waves from the upper boundary of the computational domain [6]. The most common method is an artificial absorption layer, which is placed in the vicinity of the upper boundary of the truncated domain. The parameters of such a layer are to be determined as a result of some empirical considerations, and it is usually impossible to estimate its efficiency. Another approach is the reformulation of the initial problem in semi-infinite domain to the equivalent problem in finite domain by using a transparent nonlocal boundary condition (NLBC). It is noteworthy that the NLBC is a common approach to the analogous problems in hydroacoustics and quantum mechanics [7]. For the first time, this approach was published independantly in [8] and [9]. Earth-flattening transformation yields a linear increasing modified refractive index [4] outside the computational domain. NLBC for the linear refractive index was discussed in [10] and [11]. In [12], discrete NLBC for the Crank–Nikolson scheme was derived. The derivation of NLBC for the numerical scheme leads to more stable numerical solutions. The standard Crank–Nikolson scheme is suitable in absence of terrain heterogeneities, but for the irregular terrain modeling and urban propagation, a more precise approach is required. Padé approximations for the Helmholtz equation are well suitable for scattering on horizontal obstacles [4], [13]. NLBC for Padé PE with constant refractive index outside the computational domain was considered in several studies [14], [15]. The purpose of the present study is the derivation of discrete NLBC for the Padé approximations of the Helmholtz equation with linearly increasing refractive index.

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II. Split-Step Padé Solution

Tropospheric radio-wave propagation problem is associated with the appropriate complex field component $\psi(x,z)$, defined on the semi-infinite domain. Function $\psi(x,z)$ follows the Helmholtz equation [4]

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial z^2}+k^2m^2(x,z)\psi=0 \tag{1}$$

where k is the wavenumber, $m^2(x,z)=1+2N(x,z)+2z/R$, $N(x,z)$ is the refractive index, and R is the radius of the Earth.

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It is common to use the modified refractivity M, which is related to m by

$$M = (m-1)10^6.$$

Function $\psi(x,z)$ is subject to the appropriate Earth's surface impedance boundary condition and the initial condition. By introducing the reduced function

$$u(x,z) = e^{-ikx}\psi(x,z)$$

associated with the paraxial approximation, the outgoing field can be written in the form of the one-way Helmholtz equation [4]

$$\frac{\partial u}{\partial x} = ik(\sqrt{1+L}-1)u \tag{2}$$

where

$$Lu = \frac{1}{k^2}\frac{\partial^2 u}{\partial z^2} + \left(m^2-1\right)u.$$

Step-by-step solution for (2) is given by

$$u(x+\Delta x,z) = \exp\left(ik\Delta x(\sqrt{1+L}-1)\right)u(x,z). \tag{3}$$

Following [16], we use the Padé-[n/m] approximation of the propagation operator in (3)

$$\exp(ik\Delta x(\sqrt{1+L}-1)) \approx \frac{\prod_{l=1}^{n} 1+a_l L}{\prod_{l=1}^{m} 1+b_l L}. \tag{4}$$

Coefficients a_l and b_l can be obtained by making use of the procedures, described in [17]. The accuracy of the approximation (4) affects the maximum propagation angle and the range step Δx in the numerical solution.

To solve (3) numerically, we need to discretize the operator L with respect to variable z

$$L_h u_j^n = k^{-2}D_h^2 u_j^n + \alpha_j^n u_j^n$$

where $u_j^n \sim u(n\Delta x, jh)$, $\alpha_j^n = m^2(n\Delta x, jh)$.

Discrete difference operator D_h^2 is defined as follows:

$$D_h^2 u_j^n = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{h^2}.$$

Thus, step-by-step numerical scheme can be written in the following form:

$$u_j^{n+1} = \prod_{l=1}^{p}\frac{1+a_l L_h}{1+b_l L_h}u_j^n \tag{5}$$

where $p = \max(n,m)$, $a_{p+1},\ldots,a_n, b_{p+1},\ldots,b_n = 0$.

Next, we introduce $p-1$ intermediate vectors $v_{j,1}^n, v_{j,2}^{j,n}, \ldots, v_{j,p-1}^n$ and rewrite (5) as follows:

$$\begin{cases} \left(1+b_1L_h\right)v_{j,1}^n = \left(1+a_1L_h\right)u_j^{n-1} \\ \left(1+b_lL_h\right)v_{j,l}^n = \left(1+a_lL_h\right)v_{j,l-1}^n \;\; l=2,\ldots,p-1. \\ \left(1+b_1L_h\right)u_j^n = \left(1+a_1L_h\right)v_{j,p-1}^n \end{cases} \tag{6}$$

Each row in (6) is a tridiagonal system of linear equations and can be efficiently solved numerically.

The numerical scheme (6) is stable under condition $0 \le m-n \le 2$ [18]. A stronger condition $n < m$ ensures the suppression of unwanted numerical high-frequency oscillations [4]. This regularization effect is particularly important for scattering on obstacles and irregular terrain. Note that Padé-[1/1] approximation is equivalent to the Crank–Nikolson scheme for the Claerbout PE [19].

III. Discrete NLBC

Let us denote the upper boundary of the integration domain as $z_b = Jh$. The tropospheric refractive index N does not vary significantly at altitudes more than several kilometers. Hence, it is supposed that function $N(x,z)$ vanishes when $z \ge z_b$. Operator L_h above the computational domain takes the form

$$L_h u_j^n = k^{-2}D_h^2 u_j^n + (\beta+\gamma jh)\,u_j^n.$$

By making use of the *Z*-transformation, system (6) above the computational domain may be rewritten in the following matrix form:

$$\mathbf{A}D_h^2\hat{\boldsymbol{u}}_j = k^2\left(\mathbf{B} - (\beta+\gamma jh)\,\mathbf{A}\right)\hat{\boldsymbol{u}}_j \tag{7}$$

where

$$\hat{\boldsymbol{u}}_j = \left(\,\hat{v}_{j,1}\;\ldots\;\hat{v}_{j,p-1}\;\hat{u_j}\,\right)$$

$$\mathcal{Z}\{u_j^n\} = \hat{u}_j(\xi) = \sum_{n=0}^{\infty}\zeta^{-n}u_j^n$$

$$\mathbf{A} = \begin{pmatrix} \xi b_1 & & & & -a_1 \\ -a_2 & b_2 & & & \\ & \ddots & \ddots & & \\ & & -a_{p-1} & b_{p-1} & \\ & & & -a_p & b_p \end{pmatrix}$$

$$\mathbf{B} = \begin{pmatrix} -\xi & & & & 1 \\ 1 & -1 & & & \\ & \ddots & \ddots & & \\ & & 1 & -1 & \\ & & & 1 & -1 \end{pmatrix}.$$

Consider the generalized eigenvalue problem

$$\mathbf{B} = \mathbf{APJP}^{-1},$$

where $\mathbf{J} = \mathrm{diag}\,\{s_1, s_2 \ldots s_p\}$, $\mathbf{P}$ is the matrix of right eigenvectors. Matrices $\mathbf{P}$ and $\mathbf{J}$ can be efficiently computed by making use of the generalized Schur decomposition. The existence of matrix $\mathbf{P}^{-1}$ is considered in [15].

Write down the system (7) with respect to a new variable $\hat{\boldsymbol{u}}_j = \mathbf{P}\hat{\boldsymbol{\chi}}_j$

$$D_h^2\hat{\boldsymbol{\chi}}_j = k^2\left(\mathbf{J} - (\beta+\gamma jh\mathbf{I})\right)\hat{\boldsymbol{\chi}}_j. \tag{8}$$

The system (8) consists of p independent difference equations of the form

$$D_h^2\chi_{j,l} = k^2\left(s_l - \beta - \gamma jh\right)\chi_{j,l}. \tag{9}$$

Consider first the case when $\gamma = 0$. One can easily see that the solution of each difference equation (9) can be written in the form $\chi_{j,l} = \mu_l^j$, where $|\mu_l| < 1$ is determined from the relation

$\mu_l - 2 + \mu_l^{-1} = k^2 h^2 (s_l - \beta)$. s_l are the diagonal elements of $\mathbf{J}$. Transformed NLBC will take the form

$$\hat{\boldsymbol{u}}_{J-1} = \hat{\mathbf{D}}(\xi)\hat{\boldsymbol{u}}_J$$

where

$$\hat{\mathbf{D}}(\xi) = \mathbf{P}\mathrm{diag}\left\{\mu_1^{-1}, \mu_2^{-1}, \ldots, \mu_p^{-1}\right\}\mathbf{P}^{-1}.$$

It should be noted that similar derivations of the NLBC with constant refractive index were performed in [14] and [15] for the underwater acoustics problems.

In case when $\gamma \neq 0$, difference ratio (9) represents a discrete variant of the Airy equation. Following [12], its solution can be written in the form

$$\chi_{j,l} = J_{\sigma_{j,l}}\left(-\frac{2}{\gamma k^2 h^3}\right)$$

$$\sigma_{j,l} = j - \frac{2 + (s_l - \beta)}{\gamma k^2 h^3}$$

where J is the Bessel function of the first kind. Matrix $\hat{\mathbf{D}}(\xi)$ takes the form

$$\hat{\mathbf{D}}(\xi) = \mathbf{P}\mathrm{diag}\left\{\frac{\chi_{J-1,1}}{\chi_{J,1}}, \frac{\chi_{J-1,2}}{\chi_{J,2}}, \ldots, \frac{\chi_{J-1,p}}{\chi_{J,p}}\right\}\mathbf{P}^{-1}.$$

The diagonal elements represent the ratio of Bessel functions, which can be efficiently calculated by the modified Lentz method [12].

An inverse Z-transformation yields the discrete NLBC

$$\boldsymbol{u}_{J-1} - \mathbf{D}^0 \boldsymbol{u}_J^n = \sum_{m=1}^{n-1} \mathbf{D}^m \boldsymbol{u}_J^{n-m} \tag{10}$$

where

$$\mathbf{D}^m = \frac{\tau^m}{2\pi}\int_0^{2\pi} \hat{\mathbf{D}}(\tau e^{i\varphi}) e^{im\varphi} d\varphi,\ \tau > 1. \tag{11}$$

The calculation of matrix $\hat{\mathbf{D}}(\xi)$ for a particular ξ is a time-consuming operation. For the efficient computation of the inverse transformation (11), an adaptive version of the Filon–Clenshaw–Curtis [20] quadrature was applied. This gives an opportunity to reduce the number of computations of matrix $\hat{\mathbf{D}}(\xi)$ in comparison to the Gauss quadrature. It is noteworthy that the convolution coefficients $\mathbf{D}^m$ in NLBC do not depend on the propagation environment and the antenna parameters. The latter means that the coefficients of NLBC can be calculated in advance and tabulated for use in time-critical applications.

IV. Numerical Results and Discussion

The assessment of the proposed method accuracy is carried out by the comparison with the results, obtained by the two-way SSF method, implemented in the open-source program PETOOL [21]. Artificial absorption layer is used to provide the truncation procedure in PETOOL.

The first example demonstrates the propagation in a standard atmosphere ($N(x, z) \equiv 0$). The corresponding M-profile is depicted in Fig. 1 (left). It is well known that in the standard atmosphere without refraction effects, the electromagnetic-field component decreases exponentially beyond the horizon. Thus, it is a good benchmark for the derived NLBC. The following simulation conditions were used: 3 GHz Gaussian horizontally polarized transmitting antenna with beamwidth 2° is located at the altitude of 30 m. The lower boundary condition models the sea surface. The height of the computational domain was taken equal to 300 m. Range step $\Delta x = 40$ m, altitude step h was taken as 0.1 and 0.3 m for Padé-[7/8] and SSF, respectively.

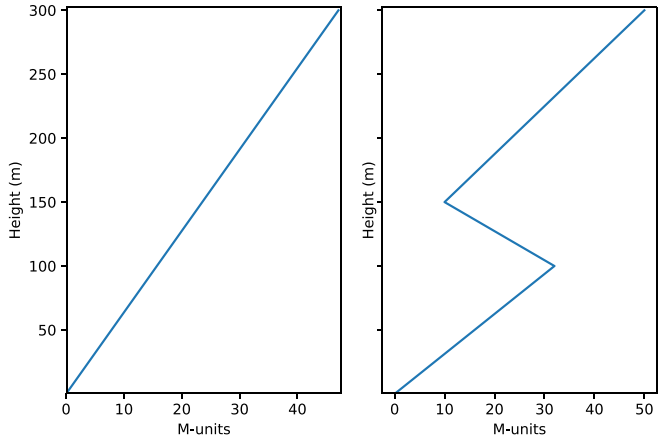


Fig. 1. Modified refractivity M-profile in a standard atmosphere (left) and elevated duct (right).

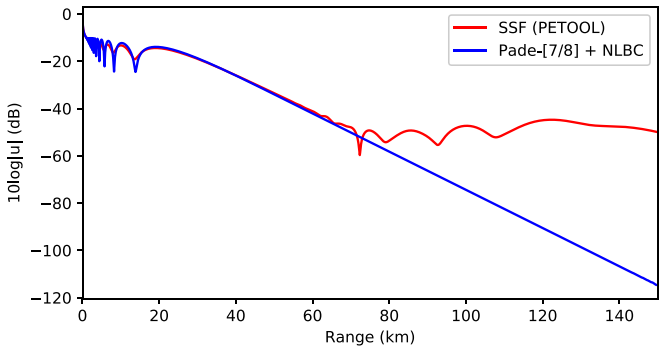


Fig. 2. Comparison of the two methods at 30 m altitude in standard atmosphere over sea surface.

It can be seen from Fig. 2 that the field amplitude calculated using the two different methods is barely distinguishable at distances less that 60 km from the source. The SSF method with the absorption layer for selected computational parameters provides an accurate solution in the dynamic range of 40 dB. SSF method yields a significant overestimation of the electromagnetic field due to false reflections from the upper boundary. The proposed truncation procedure provides an accurate solution at ranges up to 150 km. A two-dimensional distribution of the field amplitude is depicted in Fig. 3. All the computations were performed in standard double-precision arithmetic. It should be noted that it is possible to achieve a good matching of the above-mentioned methods by increasing the height of the computational domain and selecting the appropriate absorption layer in the SSF method. However, this leads to a performance degradation of the SSF method.

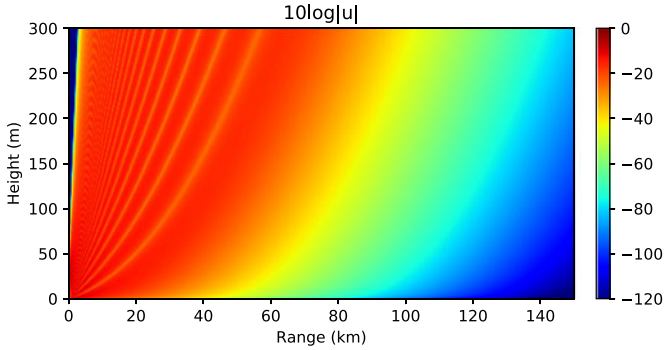


Fig. 3. Propagation in standard atmosphere over sea surface, simulated by the Padé-[7/8] scheme with NLBC.

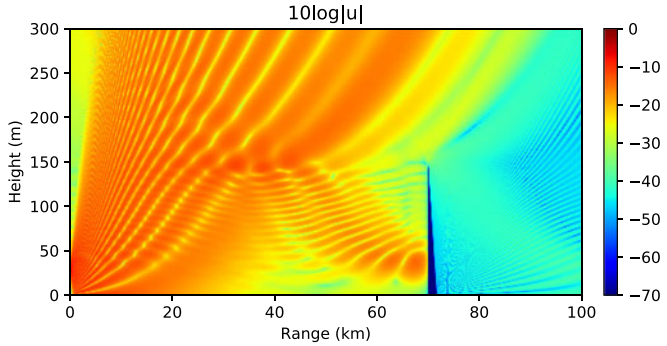


Fig. 4. Two-way modeling of the scattering on the knife edge.

The next examples deal with a canonical problem of scattering on a single perfectly conducting knife edge. It was assumed that propagation takes place in the elevated duct over a perfectly conducting surface. Refractivity M-profile for the tropospheric duct is depicted in Fig. 1 (right). Knife-edge height is 150 m, and position is 70 km in range. Other computational parameters are the same as in the previous example. In order to handle the backward scattering from the impediment, the two-way PE approach [21] was applied. The result, obtained by the proposed method, is shown in Fig. 4. The interference of the incident and reflected waves is clearly observed in front of the impediment. An incident wave is reflected by the knife edge and marched back. Shadow zone and diffraction effects are also observable behind the knife edge.

A comparison with the SSF method is depicted in Fig. 5. One can see that both the methods demonstrate a good matching. Despite using a sufficiently high-order Padé approximation, the computational costs of both the methods for the selected parameters are approximately equal.

Finally, we demonstrate the regularization properties of the applied approach in comparison with the Crank–Nikolson scheme for the Claerbout PE. This scheme is equivalent to the Padé-[1/1] PE approximation [4]. Range step Δx was taken equal to 8 m for the Padé-[1/1] scheme. It can be seen from Fig. 6 that the Crank–Nikolson scheme yields a noisy solution behind the knife edge. This effect is caused by the improper handling of the evanescent waves. It was shown in [4] that Padé approximation, where the order of the denominator exceeds that of the numerator, ensures the elimination of the boundary

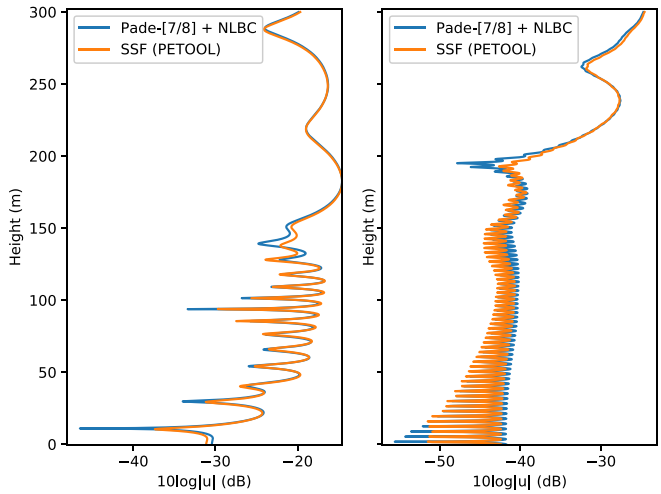


Fig. 5. Vertical field profiles at 50 (left) and 80 km (right).

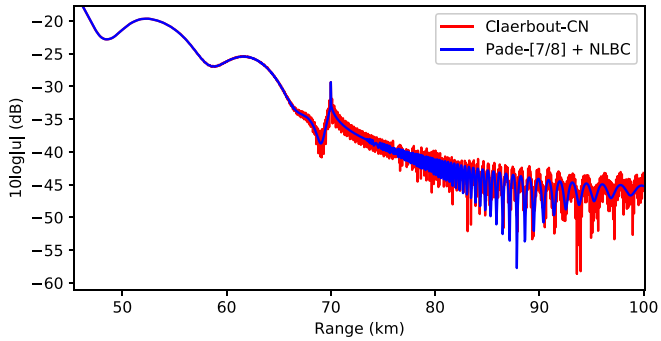


Fig. 6. Comparison of Padé-[1/1] and Padé-[7/8] schemes at 150 m altitude.

disturbances. Thus, the higher order Padé approximations are more preferable for the scattering problems.

The proposed method is implemented as an open-source Python 3 library and is freely available at [22].

V. Conclusion

It was demonstrated that the higher order Padé approximations of the Helmholtz equation are applicable for the long-range radio-wave propagation. Both effects of spatial variations of the refractive index and scattering on the impediments can be taken into account. The elaborated transparent boundary conditions provide fast and accurate truncation procedure. The proposed approach is suitable for the two-way PE method.

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